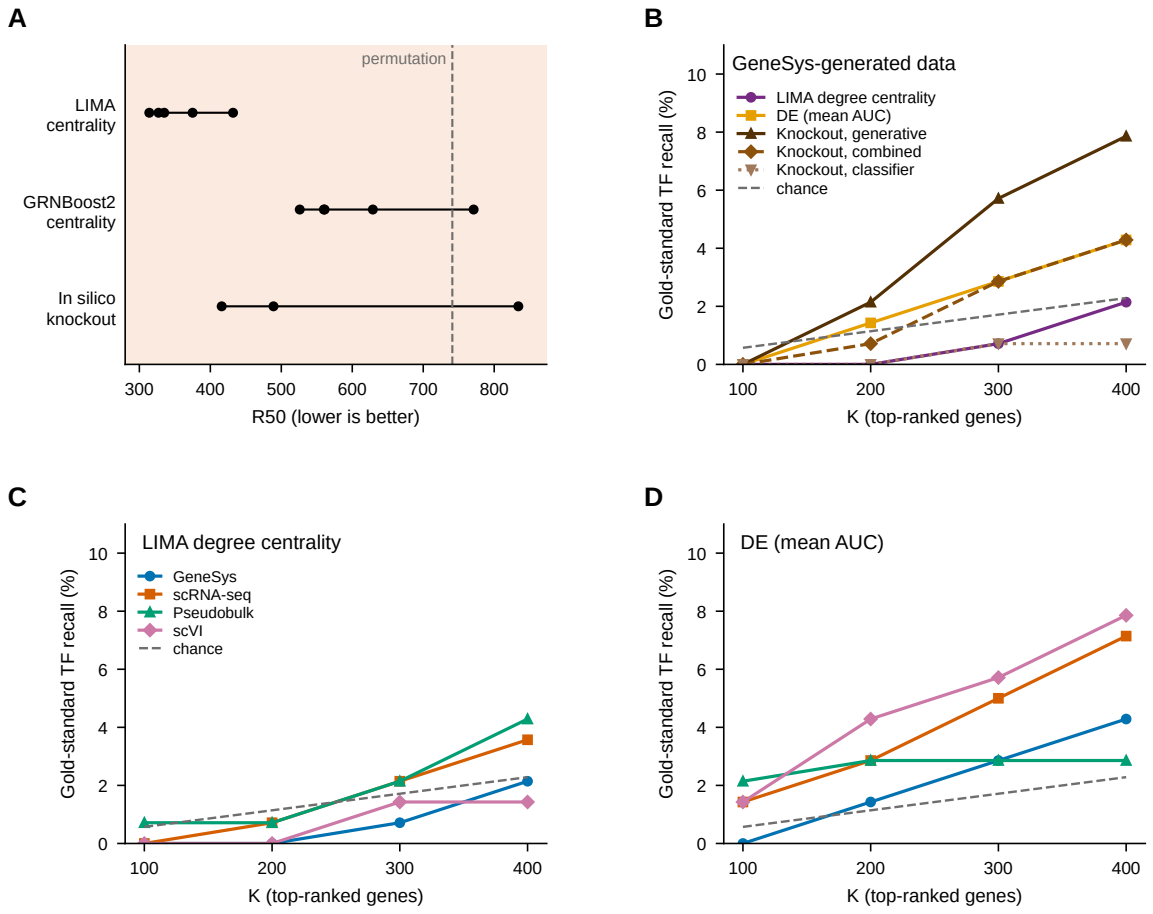
A
permutation
LIMA centrality
GRNBoost2 centrality
In silico knockout
300
400
500
600
700
800
R50 (lower is better)
B
GeneSys-generated data
LIMA degree centrality
DE (mean AUC)
Knockout, generative
Knockout, combined
Knockout, classifier
chance
Gold-standard TF recall (%)
0
2
4
6
8
10
100
200
300
400
K (top-ranked genes)
C
LIMA degree centrality
GeneSys
scRNA-seq
Pseudobulk
scVI
chance
Gold-standard TF recall (%)
K (top-ranked genes)
D
DE (mean AUC)
Gold-standard TF recall (%)
K (top-ranked genes)